

Stony Brook University
Alejandro Melo Ponce

Spring 2018

ASSIGNMENT 2

SOLUTIONS

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used in problem 4 to show your work.

1. Suppose in a survey of $N = 400$ students, 215 responded that they prefer small classes and 185 responded that they prefer large classes. Let p denote the fraction of all students who preferred small classes at the time of the survey, and $\hat{p}$ be the fraction of survey respondents who preferred small classes. (Hint: X is distributed as a Bernoulli random variable)

- Show that $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = p(1 - p)/N$.
- Use the survey result to estimate p , and calculate the standard error of your estimator. (Hint: Notice that this is the same as estimating the sample mean)
- What is the p-value for the test $H_0 : p = 0.5$ vs. $H_1 : p \neq 0.5$? Will you reject the null at the 5% significance level?
- Construct a 95% confidence interval for p .

Solution

(a)

$$E[\hat{p}] = E[\bar{Y}] = E\left[\frac{\sum_{i=1}^N Y_i}{N}\right] = \frac{1}{N} E\left[\sum_{i=1}^N Y_i\right] = \frac{1}{N} \sum_{i=1}^N E[Y_i] = \frac{1}{N} n E[Y] = p.$$

$$V[\hat{p}] = V[\bar{Y}] = V\left[\frac{\sum_{i=1}^N Y_i}{N}\right] = \frac{1}{N^2} V\left[\sum_{i=1}^N Y_i\right] = \frac{1}{N^2} \sum_{i=1}^N V[Y_i]$$

$$= \frac{1}{N^2} N V[Y] = \frac{p(1 - p)}{N}$$

where the equality $\frac{1}{N^2} V\left[\sum_{i=1}^N Y_i\right] = \frac{1}{N^2} \sum_{i=1}^N V[Y_i]$ results from the fact that under random sampling the Y_i 's are independently distributed.

(b) We have that

$$\hat{p} = \frac{215}{400} = 0.5375$$

On the other hand, we have that

$$\widehat{V(\hat{p})} = \frac{\hat{p}(1 - \hat{p})}{n} = \frac{0.5375(1 - 0.5375)}{400} = 6.2148 \times 10^{-4}$$

Thus we have that $SE = \sqrt{6.2148 \times 10^{-4}} = 0.0249$

(c) The value for the t -statistic is $t^{act} = \frac{0.5375 - 0.5}{0.0249} = 1.506$. Thus for a two-sided test we have that

$$2\Phi(-1.506) = 2\Phi(-1.506) = 2(0.066) = 0.132$$

Since the p -value is greater than the significance level, we would not be able to reject the null hypothesis at the 5% level.

- (d) Since the critical value for a 95% confidence interval is 1.96 we have that the confidence interval for p is

$$[0.5375 \pm 1.96(0.0259)] = [0.4867, 0.5882]$$

2. Let Y_1, Y_2, Y_3 and Y_4 be independent, identically distributed random variables from a population with mean μ and variance σ^2 . Let $\bar{Y} = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4)$ denote the average of these four random variables.

- (a) What are the expected value and variance of $\bar{Y}$ in terms of μ and σ^2
 (b) Now, consider a different estimator of μ :

$$W = \frac{1}{8}Y_1 + \frac{1}{8}Y_2 + \frac{1}{4}Y_3 + \frac{1}{2}Y_4$$

Show that W is also an unbiased estimator of μ . Find the variance of W .

- (c) Based on your answers to the previous questions, which estimator do you prefer $\bar{Y}$ or W ?

Solution

For this problem, recall that the four observations are independent, and identically distributed. The observations are drawn from a population with mean and variance μ and σ^2 respectively.

- (a) $E[\bar{Y}] = \mu$ and $V[\bar{Y}] = \sigma^2/4$.
 (b) $E[W] = \mu \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{2} \right) = \mu$ and $V[W] = \sigma^2 \left(\frac{1}{64} + \frac{1}{64} + \frac{1}{16} + \frac{1}{4} \right) = \frac{11}{32}\sigma^2$. Notice that since $E[W] = \mu$ the estimator is unbiased.
 (c) Both $\bar{Y}$ and W are unbiased. However, $\bar{Y}$ has a lower variance than W , thus the former is more efficient. Hence, $\bar{Y}$ is preferred.

3. You are hired by the governor to study whether a tax on liquor has decreased average liquor consumption in your state. You are able to obtain, for a sample of individuals selected at random, the difference in liquor consumption (in ounces) for the years before and after the tax. For person i who is sampled randomly from the population, Y_i denotes the change in liquor consumption. Treat these as a random sample from a $\text{Normal}(\mu, \sigma^2)$ distribution.

- (a) The null hypothesis is that there was no change in average liquor consumption. State this formally in terms of μ .
 (b) The alternative is that there was a decline in liquor consumption; state this formally in terms of μ .
 (c) Now, suppose your sample size is $n = 900$, and you obtain estimates $\bar{y} = -32.8$ and $s = 466.4$. Calculate the t statistic for testing H_0 against H_1 ; obtain the p -value for the test. (Because of the large sample size, just use the standard normal distribution.) Do you reject H_0 at the 5% level? At the 1% level?
 (d) Would you say that the estimated fall in consumption is large in magnitude? comment on the practical versus statistical significance of this estimate.
 (e) What has been implicitly assumed in your analysis about other determinants of liquor consumption over the two-year period in order to infer causality from the tax change to liquor consumption?

Solution

- (a) $H_0 : \mu = 0$.
- (b) $H_a : \mu < 0$.
- (c) The standard error of $\bar{y}$ is $s/\sqrt{n} = 466.4/30 \approx 15.55$. Therefore, the t -statistic for testing H_0 against H_a is $t = -32.8/15.55 \approx -2.11$. The one-sided p -value is given by $Pr(Z \leq -2.11) = \Phi(-2.11)$, where $Z \sim N(0, 1)$. You can find this probability in any tabulation of the standard normal distribution. The p -value is 0.0174. Because the p -value is below 0.05, we reject H_0 against the one-sided alternative at the 5% level. We do not reject at the 1% level because the p -value is greater than 0.01.
- (d) The estimated reduction, about 33 ounces, does not seem large for an entire year's consumption. Thus, even if we found statistical evidence, at the 5% level, of the reduction, it does not seem to have practical significance.
- (e) The implicit assumption is that other factors that affect liquor consumption, such as income, changes in prices due to supply shocks or preferences, are constant over the year studied.

4. On Blackboard under the Problem Set 2 folder, you can find a zip file **CPS08.zip** which contains a database **CPS08.csv** for full-time, full-year workers age 25–34, with a high school diploma or B.A./B.S. as their highest degree. A detailed description is given in the pdf file **CPS08_Description**. In this exercise, you will investigate the relationship between a worker's age (*age*) and average hourly earnings (*AHE*).

(Hint: If you are looking for general methods to save plots in R, please look at the following webpage <http://www.stat.berkeley.edu/~s133/saving.html>)

- (a) Load the dataset into R and call it **data_wage**. What command did you use? (**Do not use the "import dataset" command in Rstudio.**)
- (b) How many observations are there in your sample? How many variables?
- (c) Report the mean and the standard deviation for *AHE*. Also, create an histogram for *AHE*. Would you say that this variable is normally distributed?
- (d) Make a scatter plot of *AHE* against *age* for the entire sample. What do you observe? Do average earnings increase with age or not? Explain.
- (e) Repeat the previous question, but this time split the sample into males and females. Would you say that the relationship between age and *AHE* does not depend on gender? Explain.